

Viscosity measurement and simulation of microbubble wetting on flat surfaces with many-body dissipative particle dynamics model

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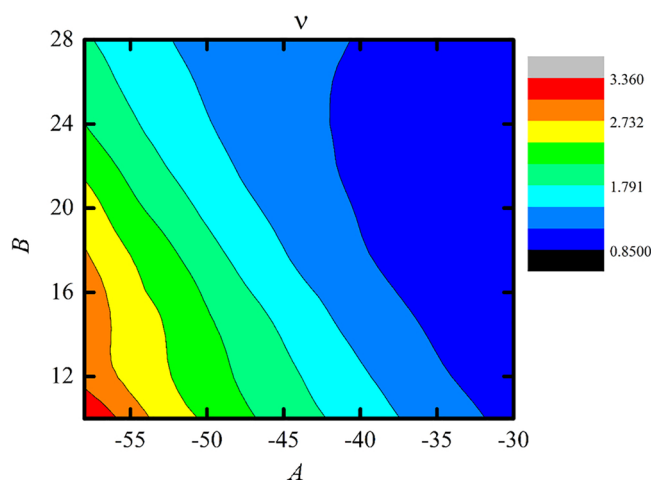
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GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Viscosity
Poiseuille flow method
Couette flow method
Many-Body dissipative particle dynamics
Bubble wetting contact angle

ABSTRACT

One of the most important flow properties of liquid is its viscosity. In the present paper, two non-equilibrium methods, Poiseuille flow method and Couette flow method, are employed to measure the viscosity of many-body dissipative particle dynamics (MDPD) model. The results obtained by these two non-equilibrium methods are consistent with each other. Based on Poiseuille flow method, we obtain a map for the viscosity of MDPD model with different numerical parameters. Furthermore, the static wetting of microbubbles is investigated by a selected system from the map. The relationship between the contact angle of bubbles and the solid-liquid interaction is obtained, which can be helpful for further microfluidic research and applications based on microbubbles.

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<https://doi.org/10.1016/j.colsurfa.2020.125559>

Received 27 June 2020; Received in revised form 6 September 2020; Accepted 6 September 2020

Available online 18 September 2020

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1. Introduction

The viscosity is an essential parameter for a fluid, which determines the flow property. If a fluid model is constructed by using a particle-based method, it is not easy to calculate the viscosity due to the “collective” characteristic [1]. Several researchers have measured the viscosity of some liquid models. Based on the micro-Poiseuille flow method, Xu et al. [2] studied the liquid argon in nanoscale in molecular dynamics (MD) simulation and found a transition which is dependent on the shear rate between the Newtonian flow and non-Newtonian flow. Backer and co-workers [1] calculated the viscosity of dissipative particle dynamics (DPD) liquid model by proposing the periodic Poiseuille flow method and compared with the shear flow method and the zero-stress method, where the former appears to be more accurate than the latter one which is an equilibrium method. Wang and co-workers [3] investigated the viscosity of a molecular dynamics model with that obtained by the pressure-driven Poiseuille flow method and presented the relationship between the viscosity increase and the load of the particle. Most of the research mentioned above focus on the Newtonian fluids, in which the non-equilibrium methods were found to be more accurate than the equilibrium method for measuring the viscosity of particle-based fluid model. For non-Newtonian fluids, such as DNA suspension and polymer fluids, the profiles of velocity and shear stress are generally approximated as a power law over a range of shear rate, for which the viscosity can also be measured by using the methods

Table 1

Comparison of the viscosity measured by PPF and SF.

ρ	S_{xz}	γ	$\eta_{\text{shearflow}}$	$\eta_{\text{Poiseuille}}$
5.8 ± 0.5	3.51 ± 0.5	0.2	17.55 ± 2.5	15.50 ± 2.8
6.8 ± 0.4	4.76 ± 0.52	0.2	23.80 ± 2.6	20.70 ± 3.5
8.0 ± 0.6	5.81 ± 0.70	0.2	29.05 ± 3.5	29.10 ± 1.6
8.7 ± 0.5	9.02 ± 1.38	0.2	45.10 ± 6.9	40.90 ± 5.3
9.6 ± 0.4	11.40 ± 1.74	0.2	57.00 ± 8.7	54.31 ± 4.6

mentioned above. Fan et al. [4] simulated DNA suspension micro-channel flows similar with the Poiseuille flow. They calculated the viscosity and implied that their results are in agreement with the experimental data, which supports that the viscosity of this type of non-Newtonian fluid can be obtained by Poiseuille flow. Fedosov and Karniadakis [5] measured the non-Newtonian viscosity of polymer fluids from plane Poiseuille flow and Couette flow. They found that the profiles of velocity satisfy the power law. Hess [6] has compared these methods and concluded that the non-equilibrium methods such as the periodic Poiseuille method and shear flow method appear more stable and accurate in MD simulations. These work illustrate that the two non-equilibrium methods can be used to measure the viscosity of a fluid in various particle-based models. As for the vapour-liquid coexistence particle based method, MDPD model has been widely used to investigate complex fluid problems including free interface. However, the

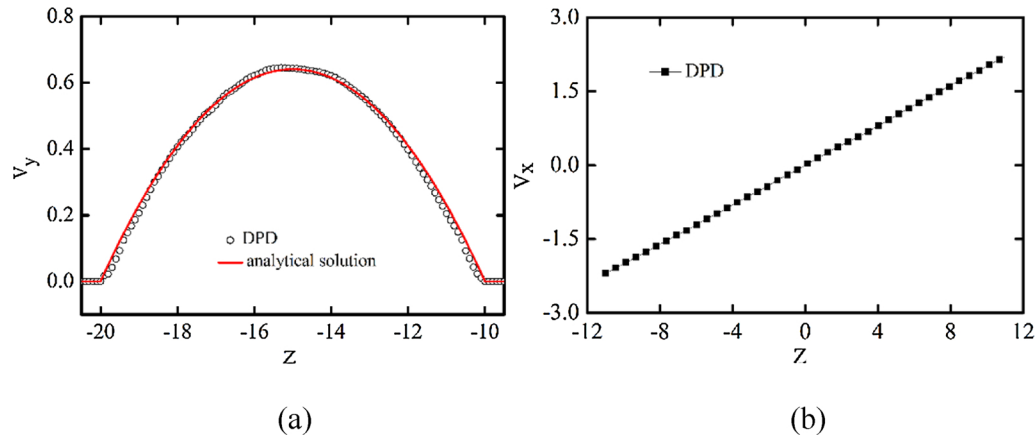


Fig. 1. Velocity profiles of the two typical flow models: (a) Poiseuille flow; (b) Couette flow.

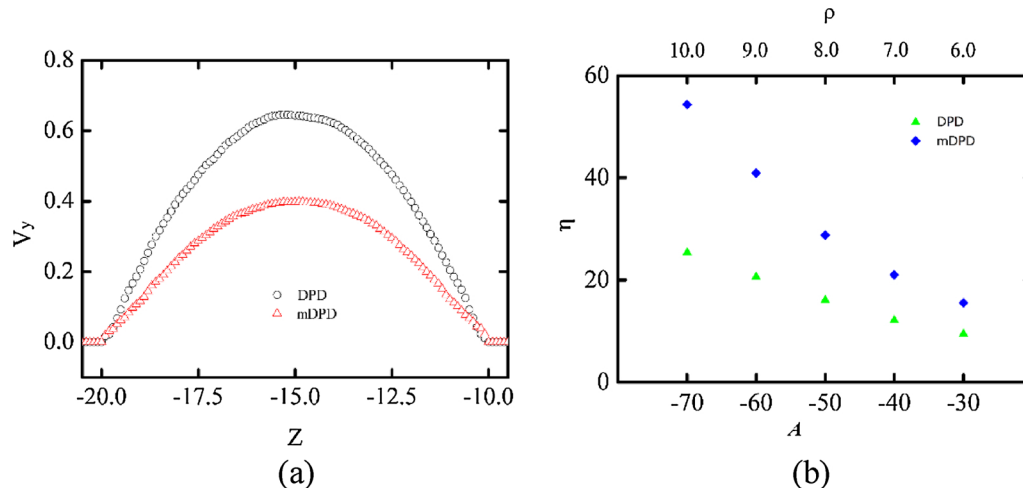


Fig. 2. Results of in DPD and MDPD models: (a) velocity profile for Poiseuille flow; (b) viscosity comparison with different attractive parameter A , under a fixed repulsive parameter.

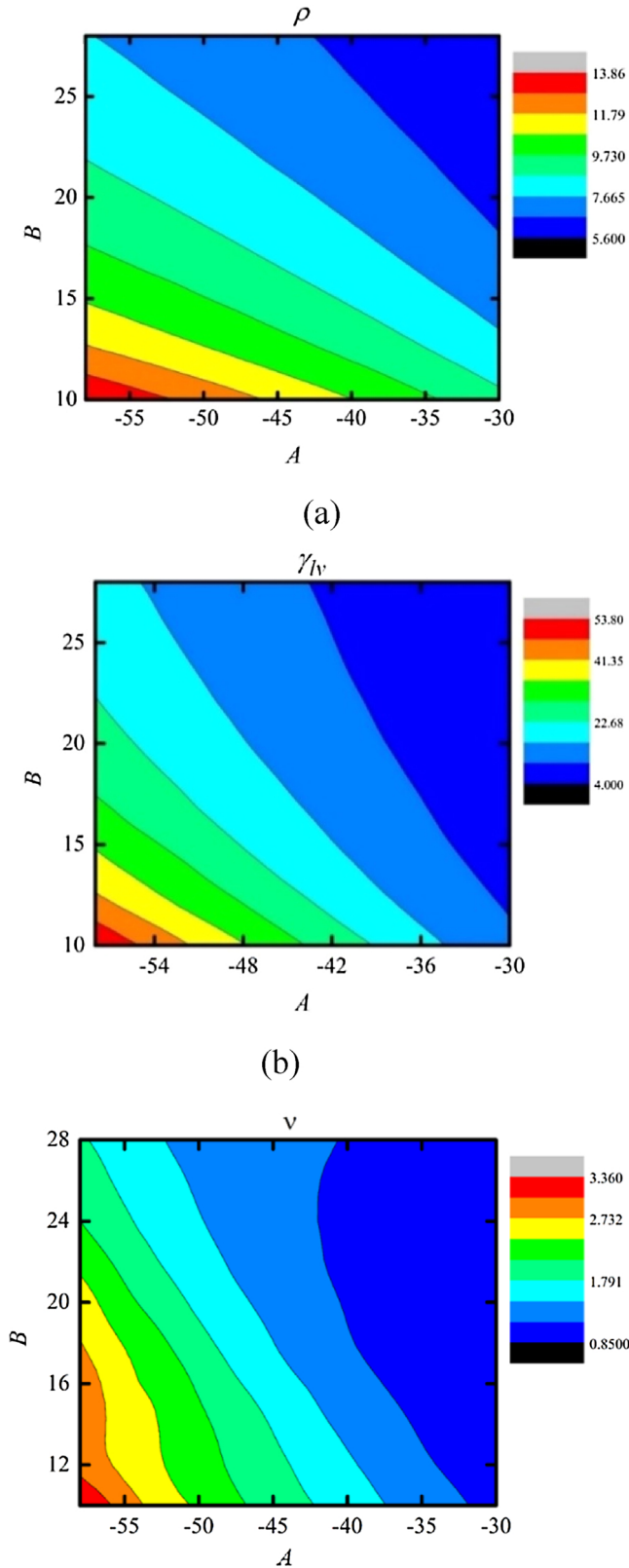


Fig. 3. Maps for the density (a), surface tension (b) and viscosity (c) of the MDPD model with various conservative parameters A and B .

viscosity of MDPD liquid model is still not very clear for further research. Zhao et al. [7] investigated the viscosity by using shear flow method and they analyzed the relationship between the viscosity and the repulsive parameter. Apart from the viscosity, the density and

surface tension, as output properties, in MDPD model also need to be computed in the simulations rather than input parameters. Thus, the specific values of them in a fluid system could be significantly different. Different from DPD model, the close-form theory to describe the basic property is still lacking in MDPD model. For convenience, The map for basic properties in MDPD model with different conservative parameters needs to be constructed as this method has been widely used to investigate microfluidic research and applications, such as droplet/bubble wetting and sliding on hydrophobic substrates.

In this paper, Poiseuille flow and Couette flow are simulated to measure the viscosity of MDPD model. In Section 2, the DPD/MDPD method and the implementation of the period Poiseuille flow and Couette flow models are introduced. In Section 3, two measurements for the viscosity of DPD and MDPD models are given. The differences between these two liquid models are analyzed in detail. Then the viscosity of MDPD model calculated by the two different methods are validated to meet with each other. Subsequently, a map of the viscosity in MDPD model correlated with different numerical parameters is provided. Finally, based on this map, the wetting process of microbubbles is investigated. The result indicates this model can be used to conduct further research and develop more applications related with microbubbles. In Section 4, we draw conclusions from the present work.

2. Method

2.1. DPD/MDPD

As a particle-based method for simulating complex fluid in mesoscale, dissipative particle dynamics (DPD) was first proposed by Hoogerbrugge and Koelman [8] in 1992 and then has been used to investigate many applications including polymer [9–11], DNA [4], and cell dynamics [12–15], etc. The motion of each particle involved the profiles of positions and velocities in DPD is controlled by Newton's second law, which is described as follows

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i, \quad \frac{d\mathbf{v}_i}{dt} = \sum_{j \neq i} \mathbf{f}_{ij} + \mathbf{F}_e \quad (1)$$

where the mass of DPD particle is normalized to 1, $\mathbf{r}_i$ and $\mathbf{v}_i$ are the position and velocity vectors of the i th particle respectively. $\mathbf{F}_e$ is the external force and $\mathbf{f}_{ij}$ is the inter-particle force between particle i and particle j , consisting of three parts:

$$\mathbf{f}_{ij} = \mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R. \quad (2)$$

The conservative force is given by

$$\mathbf{F}_{ij}^C = \begin{cases} a_{ij}(1 - r_{ij}/r_c)\mathbf{r}_{ij} & r_{ij} < r_c \\ 0 & r_{ij} > r_c \end{cases}, \quad (3)$$

where a_{ij} is the amplitude of repulsive interaction between particle i and j , $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$, $r_{ij} = |\mathbf{r}_{ij}|$ and $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$, which represents the unit vector directed from particle j to i . r_c is the cut-off radius which means the range of the interaction between two DPD particles.

The dissipative and random forces are given by

$$\mathbf{F}_{ij}^D = -\gamma\omega^D(r_{ij})(\mathbf{e}_{ij} \cdot \mathbf{v}_{ij})\mathbf{e}_{ij}, \quad \mathbf{F}_{ij}^R = \sigma\omega^R(r_{ij})\theta_{ij}\mathbf{e}_{ij}, \quad (4)$$

where the weight functions $\omega^D(r_{ij}) = [\omega^R(r_{ij})]^s = (1 - r_{ij}/r_c)^s$ at $0 < r_{ij} < r_c$ and we chose $s = 2$ here to obtain an appropriate interaction between particles. The relation between γ and σ can be described as $\sigma^2 = 2\gamma k_B T$ where $k_B T$ is the Boltzmann temperature of the system. After adding the van der Waals loop into the standard DPD by Warren [16], the modified version, named “many-body DPD”, has been used to simulate more complex fluid mechanics problems involving free surfaces with a concise form [17–26]. Here an improved conservative force can be described as

$$\mathbf{F}_{ij}^C = A\omega^C(r_{ij})\mathbf{e}_{ij} + B(\bar{\rho}_i + \bar{\rho}_j)\omega_d(r_{ij})\mathbf{e}_{ij}, \quad (5)$$

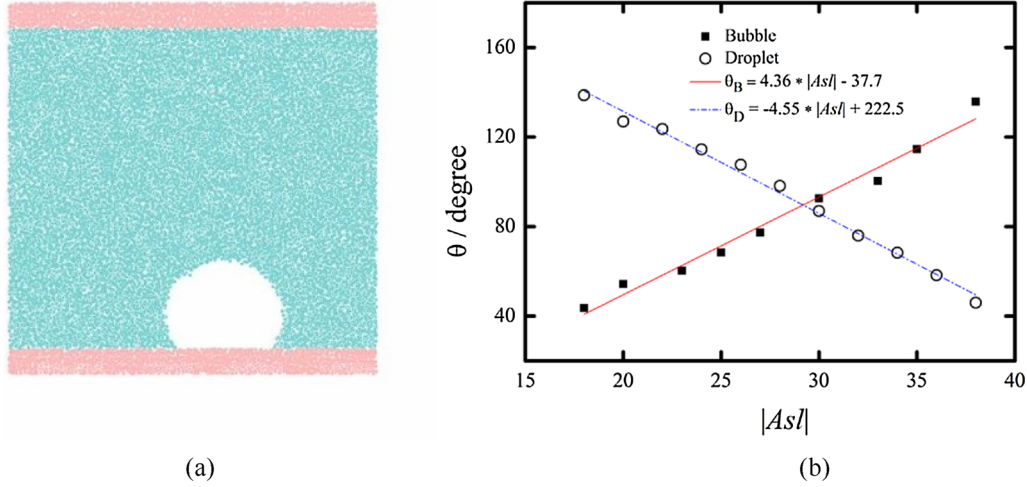


Fig. 4. Bubble wetting on flat substrates: (a) scheme of bubble wetting; (b) bubble wetting contact angle changing with solid-liquid parameter $|Asl|$.

where $A < 0$ and $B > 0$ are the amplitude of the attraction and repulsion terms, respectively. $\omega^C(r_{ij}) = 1 - r_{ij}/r_c$ and $\omega_d(r_{ij}) = 1 - r_{ij}/r_d$ are the weight functions of the attractive and repulsive interactions, respectively. $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ is the distance from particle i to particle j . The two weight functions, $\omega^C(r_{ij})$ and $\omega_d(r_{ij})$, translate the interaction between attractive form (at a long range when $r_d < r_{ij} < r_c$) and repulsive form (at a short range when $0 < r_{ij} < r_d$). Here $r_c = 1.0$ and $r_d = 0.75$ in MDPD unit as Arienti et al. [27] adopted. The local density $\bar{\rho}$ in the repulsive term is constructed by its own weight function $\omega_\rho(r_{ij})$ as follows,

$$\bar{\rho}_i(r_{ij}) = \sum_{j \neq i} \omega_\rho(r_{ij}), \quad (6)$$

$$\omega_\rho(r_{ij}) = \frac{15}{2\pi r_d^3} \left(1 - \frac{r_{ij}}{r_d}\right)^2, \quad (7)$$

and the rest two forces are the same as the original DPD model.

2.2. Poiseuille flow method (PF)

Poiseuille flow model is set by using a constant body force, as Yamada et al. did before [28]. The flow field has a parabolic velocity profile

$$v_y = \frac{\rho g_y}{2\eta} \left(\frac{L^2}{4} - z^2 \right). \quad (8)$$

In this equation, v_y is the velocity directed parallel to the body force, ρ is the mass density, g_y is the gravitational constant and η is the dynamic viscosity. L is the distance between the two plates and z is the position between the plates. Measurement of this system velocity directly gives the viscosity of the fluid system.

2.3. Shear flow method (SF)

Introducing Lee-Edwards boundary condition to DPD model is a simple method to obtain the shear flow, which has been used in the previous work [28]. One would need to move the upper and lower neighboring cells with the velocity of $V_{Cell}^{Upper} = -V_{Cell}^{Lower} = \frac{\dot{\gamma} L_y}{2}$, to obtain the shear rate of $\dot{\gamma}$ for a calculation area of (L_x, L_y, L_z) size. When the shear flow system reaches equilibrium state, one can calculate the pressure tensor by

$$P = \frac{1}{V} \left\{ \sum_{i=1}^N m_i (\mathbf{v}_i - \mathbf{u}(r_i)) \cdot (\mathbf{v}_i - \mathbf{u}(r_i)) + \sum_{j>1}^N \sum_{i=1}^{N-1} \mathbf{r}_{ij} \cdot \mathbf{F}_{ij}^T \right\}. \quad (9)$$

The stress tensor would be negative of the pressure tensor ($S = -P$), and thus the shear viscosity of a DPD/MDPD fluid can be calculated from the xz component of the stress tensor:

$$\eta = \frac{\langle S_{xz} \rangle}{\dot{\gamma}}. \quad (10)$$

The bracket in Eq. (10) represents the time average value of the shear stress.

2.4. Two basic types of flow models

Poiseuille flow and Couette flow are simulated to test the validity of the procedures. In the base case, we set parameters as $\rho = 5.8$, $\gamma = 8.0$, and $k_B T = 1.0$ to define the type of liquid. An external force of $g = 0.1$ is applied to drive the liquid particles in Poiseuille flow model, while a static shear rate $\dot{\gamma} = 0.2$ is adopted to define the shear flow velocity amplitude. In the first model, the measured velocity profile shown in Fig. 1 (a) agrees with Eq. (8) and in the second one, the velocity profile shows a linear distribution with constant slope (represents the shear rate $\dot{\gamma}$), as shown in Fig. 1 (b).

3. Simulation results and discussion

3.1. Comparison of the viscosity in DPD and MDPD models

The viscosity related with different parameters in DPD and MDPD models are calculated in further. The profiles of velocity and viscosity are shown in Fig. 2. In Fig. 2 (a), the number density is also set as 5.8, and the external force is 0.1. The results show that the velocity profile and the viscosity in MDPD model are significantly different with that in DPD model. To compare the viscosity with that of DPD, we set the density to be the same in DPD, which is related with the attractive parameter in MDPD. More detailed relation between the basic properties and the parameters in MDPD can be seen in the subsection 3.3. Under the same density and temperature, the maximum of velocity in MDPD model is around half of that in DPD model. And then we test a series of measurements with different conservative parameter A , and the results are shown in Fig. 2 (b). Correspondingly, the viscosities in these MDPD systems are much larger than those of DPD model. From our simulation results, the viscosity decreases with the attractive interaction A , which is much faster than that in DPD model.

3.2. Comparison of the viscosity measurement in MDPD model

The viscosity of MDPD model with different number densities by

using Poiseuille flow and Couette flow are compared. The temperature of the system is the same as that in the previous tests. At this state point, the results are as follows:

The average viscosity of the two methods are nearly in agreement with each other. As the theoretical formula for calculating the viscosity of MDPD model is still lacking, we measured the collective values by using two non-equilibrium methods instead, which could be adopted in a reasonable range for further research. To obtain them with more accuracy, we tested every variable for ten times with different initial random numbers in the simulations. There exist the errors between different tests even the sizes of samples for calculating the results are the same, which can be understood as statistical error (Table 1).

3.3. Maps for basic properties in MDPD model with different conservative parameters

By adopting Poiseuille flow method, the maps for density, surface tension and viscosity in MDPD model with different conservative parameters, i.e. A and B are obtained, as shown in Fig. 3 (a–c). While the dissipative coefficient parameter is set as a constant value $\gamma = 18.0$. The systems are simulated with the attractive parameter A varying from -58 to -30 combined with the repulsive parameter B ranging from 10 to 28, and there are 150 groups of test in total. The results show that both the density and surface tension have a linear relationship with either the attractive or the repulsive parameter. While the viscosity is significantly different. It is known that in the classical DPD, the viscosity is dependent on the density and the dissipative parameter, meanwhile, is independent on the conservative parameter. While in MDPD model, the local number density of fluid particles also relies on the conservative force, and then the equation of state can describe a “many-body” system with free interface. Based on this, the conservative parameters can directly control the density of the liquid system, and consequently affect the shear viscosity. Therefore, the viscosity is not only relevant with the dissipative parameter any more. As shown in Fig. 3 (c), the map shows that the kinematic viscosity ($\nu = \eta/\rho$) changes with the conservative parameters A and B , as a nonlinear relationship. For a constant value of A , it can be seen that the viscosity generally decreases with increasing B , which is in agreement with that in Zhao et al.’s work [7]. Although this map cannot cover all parameters in MDPD model, it can provide a useful concept for further research with various systems. The magnitude of surface tension is determined by the interactions between MDPD particles, which is an output property of the MDPD system rather than an input parameter. The exact value of the surface tension needs to be computed from MDPD simulations. There are several methods which can be applied to compute the surface tension of a MDPD fluid. The first one is to compute the pressure difference between the inside and outside of a droplet for various radii. And then the surface tension can be obtained according to the Young-Laplace equation, i.e. $\Delta p = p_{in} - p_{out} = 2\sigma_{lv}/R$, as mentioned Ref. [27]. The second one is the pendant drop method which can calculate the surface tension by fitting the shape of a pendant droplet and then compare with the theory, as mentioned in Ref. [16]. This method is widely used in experimental tensiometers but not suitable for micro/nanodroplet in which the gravity is negligible. The third method is the easiest one that needs a thin film in a periodic computational box. The surface tension can be calculated by testing the stress tensor in the Irving-Kirkwood expression, which is described as $\sigma_{lv} = \int \left[p_{zz} - \frac{1}{2}(p_{xx} + p_{yy}) \right] dz$, where p_{xx} , p_{yy} , and p_{zz} are the three diagonal components of the pressure tensor, with x and y parallel to the planar interface and z normal to it. In our work, we adopted the third method to calculate the surface tension.

3.4. Microbubble wetting on flat surfaces in MDPD model

Based on the maps given by the above subsection, a system is chosen to simulate the static wetting of microbubbles on flat hydrophobic

substrates with different wettabilities, as shown in Fig. 4 (a). The conservative parameters in the model are $A = -40$, $B = 25$. The physical units corresponding to the dimensionless units in the system are obtained by referring to our previous work [29]. The static contact angles of microbubble wetting on flat substrates with different wettabilities are plotted in Fig. 4 (b). The results are compared with that of droplets wetting on the same surfaces. The static contact angles are obtained by calculating the height and wetting length of the bubbles, which has been used to measure the contact angles of droplets in previous work [30–32]. From our simulations, the fitting formula to predict the contact angle of microbubble wetting on flat surfaces is obtained, i.e. $\theta_B = 4.36|A_{sl}| - 37.7$, which presents a linear relationship between the static contact angle and the solid-liquid interaction. While a similar one of droplets wetting on flat surfaces is also provided as $\theta_D = -4.55|A_{sl}| + 222.5$, which is close to that given by Li et al. [25]. Comparing to that of droplets wetting on flat surfaces, the values of bubble contact angle experience a reversely linear monotony. The summation of contact angles measured the bubbles and droplets is equal to 180 degree, which is in good agreement with that in the work by Wu et al. [33]. It is worth noting that the size and line tension effect on the contact angle are neglected as the size of the bubble is larger than ~ 10 micrometers, for which Zhang et al. [34] implied that the bubble shape could be different under this critical value.

4. Conclusion

The viscosity of MDPD model is measured by using two typical fluid models, i.e. Poiseuille flow and Couette flow. By comparing the measured viscosity in these two flow models, it shows that the measurement results are in consistent with each other, which can validate the rationality of the results as the relevant theory is still lacking. Based on Poiseuille flow model, a map of viscosity related to different numerical parameters is provided, which can be used to easily obtain basic flow properties for different liquid systems. It is noted that although the map in present work does not involve all numerical parameters in MDPD model, this concept can also be extended to other systems. Moreover, a selected system from the map is adopted to simulate the wetting contact angle. The summation of bubble and droplet contact angles is around at a reasonable value of 180 degree. The fitting formula to predict the bubble contact angle is also obtained. In present work, the relation between the viscosity measurement part and the bubble model part is not so apparent as the bubble wetting in this part is static. However, in dynamic process, such as bubble impacting on surface with various We numbers under different Re numbers, the viscosity could affect the process of impacting and retraction, especially at micro-/nanoscale. Also, the change of liquid-gas interfacial tension indeed affects the dynamics of the bubble like nucleation and growth. This work can provide insights on the properties of MDPD model and be helpful to do further microfluidic research and applications related with bubbles.

CRedit authorship contribution statement

Lanlan Xiao: Data curation, Visualization, Investigation, Writing - original draft. **Kaixuan Zhang:** Conceptualization, Methodology, Software, Writing - review & editing. **Jiayi Zhao:** Software, Validation, Writing - review & editing. **Shuo Chen:** Conceptualization, Methodology, Writing - review & editing. **Yang Liu:** Supervision, Writing - review & editing.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be

construed as influencing the position presented in, or the review of, the manuscript entitled.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant Nos. 11872283, 11902188) and Shanghai Science and Technology Talent Program (Nos. 19YF1417400, 20YF1432800).

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